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Reading Temperature & Humidity on Nano 33 BLE Sense Rev2

Learn how to measure and print out the humidity and temperature values of your surroundings using the Nano 33 BLE Sense Rev2

Author · Jose Garcia · Last revision · 10/31/2025

In this tutorial we will use an **Arduino Nano 33 BLE Sense Rev2** board to measure and print out the humidity and temperature values of your surroundings, made possible by the embedded **HS3003** sensor.

Goals

The goals of this project are:

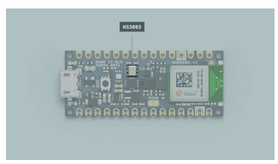
- Learn how to output raw sensor data from the Arduino Nano 33 BLE Sense Rev2.
- Use the HS300x library.
- Print temperature and humidity values in the Serial Monitor when they are within a certain range.
- Create your own temperature and humidity

Help

Hardware & Software Needed

- Arduino Nano 33 BLE Sense Rev2.
- This project uses no external sensors or components.
- IDE 2

The HS3003 Sensor



The HS3003 sensor.

Temperature sensors are components that convert physical temperature into digital data. Likewise, humidity sensors are able to measure atmospheric moisture levels and translate that into electrical signal. As such, temperature and humidity sensors are essential for environmental monitoring especially in and around sensitive electronic equipment.

The HS3003 is an ultra-compact sensor for relative humidity and temperature. We will use the I2C protocol to communicate with the sensor and get data from it. The sensor's range of different values are the following:

- Humidity range: 0 to 100 %
- Temperature accuracy: ± 0.5 °C,15 to +40 °C
- Temperature range: -40 to 120°C

These types of sensors are used more than you would think and are found in various everyday objects!

Some of the useful ways they are used are seen in the following applications:

- Air conditioning, heating and ventilation.
- Air humidifiers.
- Refrigerators.
- Wearable devices.
- Smart home automation.
- Industrial automation.
- Respiratory equipment.
- Asset and goods tracking.

If you want to read more about the HS3003 sensor you can take a look at the datasheet

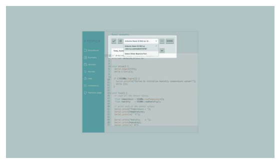
Creating the Program

1. Setting up

Let's start by opening the Arduino Cloud Editor, click on the **Libraries** tab, search for the **HS300x** library, then in **Examples**, open the **ReadSensors** example. Once the sketch is open, rename it as **Temp_Humidity**.

2. Connecting the board

Now, connect the Arduino Nano 33 BLE Sense Rev2 to the computer to check that the Cloud Editor recognises it, if so, the board and port should appear as shown in the image. If they don't appear, follow the instructions to install the plugin that will allow the Editor to recognise your board.



Selecting the board.

3. Printing temperature and humidity values

Now we will need to modify the code of the example in order to print the temperature and humidity values only when 0,5ºC degree has changed or the humidity has changed at least 1%. To do so, let's initialize the following variables before the `setup()` function.

```
float old_temp = 0;
float old_hum = 0;
```

In the `setup()`, the `HS300x.begin()` function inside an `if` statement will print a message, as a string, in case the sensor has not been properly initialized.

Then, in the `loop()`, let's modify the example code by code after the `HTS.readTemperature()` and `HTS.readHumidity()` functions respectively.

```
// check if the range
  // and if the range
  if (abs(old_temp -
  {
    old_temp = temper
    old_hum = humidit
    // print each of
    Serial.print("Tem
    Serial.print(temp
    Serial.println("
    Serial.print("Hum
    Serial.print(humi
    Serial.println("
    Serial.println();
  }
```

With this part of the code, we will print out the temperature and humidity values when the temperature increases or decreases more than 0,5ºC , or, when the humidity values change up or down more than 1%.

4. Complete code

If you choose to skip the code building section, the complete code can be found below:

```
/*
  HS300x - Read Sens

  This example reads
  Nano 33 BLE Sense
  values to the Seri

  The circuit:
  - Arduino Nano 33

  This example code
*/

#include <Arduino_HS

float old_temp = 0;
float old_hum = 0;

void setup() {
  Serial.begin(9600)
  while (!Serial);

  if (!HS300x.begin(
    Serial.println("
    while (1);
  }
}
```

Testing It Out

After you have successfully verified and uploaded the sketch to the board, open the Serial Monitor from the menu on the left. You will now see the new values printed. If you want to test out whether it is working, you could slightly breathe (exhale) on your board and watch new values when the humidity, as well as the temperature, levels rise or decrease. The following image shows how the data should be displayed

Temperature & humidity printed in the Serial Monitor.

Troubleshoot

Sometimes errors occur, if the project is not working as intended there are some common issues we can troubleshoot:

- Missing a bracket or a semicolon.
- If your Arduino board is not recognised, check that the Create plugin is running properly on your computer
- Accidental interruption of cable connection.

Conclusion

In this simple tutorial we learned how to read temperature and humidity values from the **HS3003** sensor using the HS300x library, and how to use the sensor embedded in the Arduino Nano 33 BLE Sense Rev2 board, to measure and print out humidity and temperature values from the environment.

Was this article helpful?